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Additive delivery systems and containers

Abstract

A compact filtering and additive delivery system which is readily adaptable to a portable container, such as a sports bottle, and receives a modular additive container for the metered delivery of additive, such as flavor concentrate, to a stream of base liquid as the base liquid is drawn or dispensed from the container. The modular additive container configuration on the delivery system allows a consumer/user to experience different additives, such as different flavors or supplement compositions, for a given supply of base liquid, such as water, stored in the container. The system may be readily used with off-the-shelf containers, such as disposable water bottles. An additive container configuration provides modular additive delivery system as described herein.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation and claims the priority benefit of U.S. patent application Ser. No. 17/539,833, filed on Dec. 1, 2021. This application is a continuation and claims the priority benefit of U.S. patent application Ser. No. 15/791,348, filed on Oct. 23, 2017, which is a continuation and claims the priority benefit of U.S. patent application Ser. No. 14/181,354, filed on Feb. 14, 2014, of the same title, now U.S. Pat. No. 9,795,242, which claims the priority benefit of U.S. Provisional Patent Application Ser. No. 61/764,861, titled “BOTTLE WITH REMOVABLE FLAVOR CARTRIDGE IN WHICH FLAVOR CONCENTRATE MIXES WITH WATER AS USER DRINKS,” filed on Feb. 14, 2013. The disclosures and subject matter of all of the above-described applications is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(1) The disclosure relates to dispensing and delivery systems for beverages and other products. More specifically, the disclosure relates to dispensing and delivery systems for providing an additive, such as flavoring or supplements, to a base liquid, such as water, as the base liquid is dispensed from a container. The disclosure also relates to containers for containing and storing an additive to be used in additive delivery systems.

2. Prior Art

(2) The prior art includes various devices for providing additives to a base liquid. Such devices include pre-mix systems, such as those exemplified in U.S. Pat. No. 7,306,117, in which a predetermined amount of additive is dispensed into a base liquid within the container and mixed therewith prior to consumption.

(3) Prior art systems also include additive delivery systems in which an additive is dispensed as a base fluid is dispensed from a container. Such delivery systems are exemplified by U.S. Pat. No. 8,230,777, which describes a dispensing system in which a base liquid flows through a supplement area containing solid supplements, and U.S. Pat. No. 8,413,844, which describes a water dispenser (pitcher) having a filter and an additive chamber in which the additive is dispensed as water is poured from the dispenser.

(4) Such prior art systems, however, suffer from the drawbacks of requiring rather elaborate and relatively expensive dispensing systems and fail to combine filtration and additive delivery functionality in a manner that is readily adapted to portable, compact, portable containers, such as sports bottles. There is thus a need for additive delivery systems and additive containers that are readily adaptable to portable base liquid containers and which address the aforementioned and other needs in the art.

SUMMARY OF THE INVENTION

(5) One aspect of the invention provides a compact filtering and additive delivery system, which is readily adaptable to a portable container, such as a sports bottle. The additive delivery system

provides filtration and additive delivery to a base liquid while maintaining a generally consistent flow direction of the base liquid. The additive delivery system may receive a modular additive container selected by the user and provides for the metered delivery of additive, such as flavor concentrate, to a stream of base liquid as the base liquid is drawn or dispensed from the bottle. The modular additive container configuration on the delivery system allows a consumer/user to experience different additives, such as different flavors or supplement compositions, for a given supply of base liquid, such as water, stored in the container.

(6) Another aspect of the invention provides a compact filtering and additive delivery system that may be used with off-the-shelf containers, such as disposable water bottles.

(7) Yet another aspect of the invention provides a container configuration that is suitable for a modular additive delivery system as described herein.

(8) Yet another aspect of the invention provides for uniform mixing of additive to a base liquid as the mixture is consumed, as well as a substantially uni-directional flow of base liquid and mixed additive/base liquid composition from the container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other attendant advantages and features of the invention will be apparent from the following detailed description together with the accompanying drawings, in which like reference numerals represent like elements throughout. It will be understood that the description and embodiments are intended as illustrative examples and are not intended to be limiting to the scope of invention, which is set forth in the claims appended hereto.

(2) FIG. 1 is an exploded isometric view of an additive delivery system, combined with a sports bottle and filtration system according to an aspect of the invention;

(3) FIG. 2 is a detailed exploded isometric view of an additive delivery system according to an aspect of the invention;

(4) FIG. 3 is a cross-sectional exploded view of the additive delivery system of FIG. 2;

(5) FIG. 4 is an isometric view of an additive container according to an aspect of the invention;

(6) FIG. 5 is a cross-sectional view of an assembled additive delivery system according to an aspect of the invention;

(7) FIG. 6 is a cross-sectional view showing flow paths of an additive delivery system according to an aspect of the invention.

DETAILED DESCRIPTION

(8) Referring to FIG. 1, an additive delivery system **100** according to an aspect of the invention includes an upper cap assembly **120** and a lower cap assembly **140**, in combination with a sports bottle **10**. Sports bottle **10** may include an inner volume for containing a base fluid, such as water, and a threaded mouth **12** for receiving and sealingly engaging the lower cap assembly **140**. As will be described in more detail below, upper cap assembly **120** and lower cap assembly **140** cooperate to house a removable additive module **200**.

(9) Referring additionally to FIG. 2, upper cap assembly **120** may include a spout **121** and a re-sealable spout closure **122** that cooperates therewith in a known manner. Upper cap assembly may also include an upper cap **123** having threaded interior wall **125** and an additive module enclosing wall **127** defining an additive module receiving space **128** for receiving additive module **200**.

(10) Lower cap assembly **140** may include a lower cap **141** having a container engaging female threaded portion **143** and an outer surface with gripping recesses formed therein. A lower additive module enclosure **150** cooperates with the lower cap **141** and upper cap assembly **120** to house and enclose additive module **200**, as will be described below. Lower cap assembly **140** may include a filter assembly **170**, which houses a filter for filtering the base liquid. A delivery tube **155** extends

to the bottom of the base liquid container and provides for the travel of base liquid through the filter assembly **170**, lower cap assembly **140**, additive module **200** and, ultimately, to spout **121**, as will be described in more detail below.

(11) Referring additionally to the cross-sectional exploded view in FIG. **3**, the additive module enclosing wall **127** of upper cap assembly **120** may include a retaining lip formed therein for engaging a retaining groove **214** on additive module **200**. Lower cap assembly **140** may include an upper cap engaging male threaded portion **142** for engaging the female threads **125** on the upper cap **123**. As will be recognized, this configuration allows easy removal and replacement of the additive module by the consumer as the upper cap assembly **120** may be unscrewed and removed with the additive module **200** remaining secured on the upper cap assembly **120** by way of the retaining lip **129** and retaining groove **214**, which provide a snap fitting of the additive module **200** to the upper cap assembly. In addition, during removal of upper cap assembly **120**, the lower cap **141** may remain secure on the container, preventing contamination or spillage of the base liquid.

(12) According to an aspect of the invention, for safety and sanitary purposes, the additive module **200** may be provided with a transparent safety enclosure cap, which may engage the cap threads **125**, and enclosed the additive module **200**. In this configuration, the spout **121**, spout closure **122** and safety enclosure cap (and thus the additive module, snap fit within the cap) may be sealed within a frangible thermoplastic membrane for consumer safety and product freshness. The transparent safety cap allows consumers to view the details/branding of the additive module before purchase, for example.

(13) Lower cap **141** may also be provided with an annular wall **144** for engaging and fitting within a correspondingly-shaped upper annular wall **151** on the lower additive module enclosure assembly **150**. Annular wall **144** may be provided with a retaining groove **145** for receiving a cooperating retaining lip **154** formed on the interior of upper annular wall **151** of the lower additive module enclosure assembly **150**. A lower annular wall **152** and bottom wall **153** define a space for receiving a lower portion of the additive module **200**. One or more additive module membrane piercing projections **180** may extend upward from the bottom wall **153** to pierce a frangible membrane **212**.

(14) Tube **155** extends upward through the lower module enclosure assembly **150** to permit flow of the base liquid. One or more metering ports **157** may be formed in the tube **155** near the bottom wall **153** to permit flow of additive concentrate from the interior of lower additive module enclosure **150** to the interior of tube **155** by venturi effect as the base liquid is drawn through tube **155**. One or more mixing projections **156** may extend within the interior of tube **156** to induce turbulent flow and thereby mix the additive concentrate with the base fluid.

(15) Filter assembly **170** may include a filter housing **172**, which is snapped in place on a corresponding filter housing top **174** using a lip and groove retainer. A one-way check or flapper valve **160** formed of silicone rubber may be provided at the inlet of tube **155** from the interior of filter housing **172** to prevent backflow of the base fluid into the filter housing **172**. An active carbon filter element (not shown in FIG. **3**) may be provided on the interior of filter housing **172**, as will be described in more detail below.

(16) Referring additionally to FIG. **4**, an additive module **200** according to aspects of the invention may include a generally cylindrical shape having an outer wall **204** provided with a retaining groove **214** formed therein and extending to a top wall **205** having a conical sealing surface **208**. Conical surface **208** is adapted to sealingly engage a correspondingly shaped flexible seal **130** (FIG. **3**), which may be made of silicone rubber or other material, and provided in upper cap assembly **120**. An interior annular wall **206** defines, with upper wall **205** and outer wall **204**, an interior additive containing space **210** for containing an additive, such as flavor or supplement concentrate. A frangible lower membrane **212**, which may be made of foil or other material, forms a lower wall to contain and seal the additive within the module. As will be recognized, the module configuration provides for a consumer to carry and preserve a number of different additives for use

with the additive delivery system and a single supply of base liquid in a container, such as a sports bottle.

(17) FIG. 5 illustrates a cross-section of an assembled additive delivery system according to an aspect of the invention. Spout closure **122** is shown in a closed position. Upper cap **123** is shown in an engaged position (i.e., screwed on) on the male threaded portion **142** of the lower cap **141**, with the additive module enclosing wall **127** of the upper cap assembly **120** extending downward into the annular opening of the lower cap **141** and securing the additive module **200** therein. The upper annular wall **151** of lower additive enclosure assembly **150** engages the module holder receiving wall **144** of the lower cap assembly **140** via lip **154** and groove **145** to provide a snap fitting. Upper cap assembly **120**, lower cap assembly **140** and lower additive module enclosure assembly thereby cooperate to provide a sealed containing space for the additive module **200**.

(18) Tube **155** extends upward within the inner tube formed by inner wall **206** of additive module **200**. The conical sealing surface **208** is engaged by a correspondingly shaped silicone seal **130** secured within the upper cap assembly **120**. In this position, the frangible membrane **212** of additive module **200** would be pierced or ruptured by piercing projections **180** (FIG. 3) and additive concentrate stored within additive module **200** would flow into the space **158**. FIG. 5 also shows a cylindrical active carbon filter element **175** housed within the filter housing **172** to filter the base fluid.

(19) Referring additionally to FIG. 6, the base fluid is filtered as it flows from the lower portion of tube **155** to an interior space **178** defined by filter element **172**, through the check valve **160**, and upward into the upper portion of tube **155**. Owing to a venturi effect, additive concentrate, represented by arrows "A" is drawn from the space **158**, through metering ports **157** and into the interior of tube **155** where additive is mixed with the base fluid. Applicants have discovered that metering ports having a diameter of approximately 0.03 inches provide for suitable metering of additive liquid, while permitting the retention (non-leakage) of additive liquid from the additive module when flow of the base liquid is not occurring. As will be recognized, the diameter of metering ports may be varied depending on the viscosity of the additive liquid and other parameters such that flow occurs when needed but not when the base liquid is not flowing in the delivery tube. Mixing projection **156** enhances the mixing and uniform dilution of the additive within the base fluid. The mixed additive and base fluid composition continues up the tube **155** through the spout

(20) As will be recognized, the flow of fluid through the system may be facilitated by suction provided by the consumer or by the squeezing of the container, which may be made of a suitably flexible material, or by both. As will also be recognized, flow from the base liquid container, through the filter assembly, delivery tube and additive module is substantially in a single, linear direction, without diversion, thereby providing for efficient flow of base liquid and mixed additive/base-liquid composition from the container and providing a configuration that is particularly adaptable to a sports bottle or other compact, portable, handheld container.

(21) As will also be recognized, the additive delivery system may be used with standard, disposable water or beverage bottles through suitable adaptation of the fastening implements on the lower cap assembly.

(22) As will also be recognized, suitable thermoplastic polymers may be used to form the various aforementioned elements, including polyethylene terephthalate (PET), polycarbonate, high-density polyethylene (HDPE) and others.

(23) It should be understood that implementation of other variations and modifications of the invention in its various aspects may be readily apparent to those of ordinary skill in the art, and that the invention is not limited by the specific embodiments described herein. It is therefore contemplated to cover, by the present invention any and all modifications, variations or equivalents that fall within the spirit and scope of the claims that follow.

Claims

1. An additive delivery system for use with a portable liquid container configured to hold a base liquid, the system comprising: a base liquid container having an inner volume to hold a base liquid and a threaded mouth; an additive reservoir including at least one wall defining a space holding a liquid additive and an outlet from which the liquid additive exits the additive reservoir; a cap assembly having an upper cap portion and a lower cap portion, the lower cap portion being configured to threadedly engage with the threaded mouth, and the upper cap portion including a spout configured for movement between open and closed positions to control flow of liquid from the spout, and an upper component to which the additive reservoir is attached, the upper component configured to threadedly engage with the lower cap portion to secure the additive reservoir to the lower cap portion and to threadedly disengage from the lower cap portion permit removal of the upper cap portion and the additive reservoir from the lower cap portion; and a base liquid flow path extending through the cap assembly to the spout, the base liquid flow path fluidly coupled to receive the liquid additive from the additive reservoir at a mixing area to mix the liquid additive with the base liquid and including a one-way valve in the base liquid flow path upstream of the mixing area that prevents backflow of base liquid into the base liquid container.
2. The system of claim 1, the cap assembly further comprising a lower component that at least partially defines an additive reservoir enclosure in which the additive reservoir is held, and a metering port configured to receive the liquid additive from the outlet of the additive reservoir and provide the liquid additive to the mixing area.
3. The system of claim 2, the lower component including an opening to permit the base liquid from the inner volume to flow into the additive reservoir enclosure.
4. The system of claim 2, wherein the lower component includes an opening to permit the base liquid from the inner volume to flow into the additive reservoir enclosure, and the base liquid that flows into the additive reservoir enclosure through the opening of the lower component does not contact the liquid additive in the additive reservoir.
5. The system of claim 1, wherein the upper cap portion is configured to be removed from the lower cap portion while the lower cap portion is engaged with the base liquid container.
6. The system of claim 5, wherein the upper component includes a female threaded part configured to threadedly engage with a male threaded part on the lower cap portion.
7. The system of claim 1, wherein the additive reservoir and the upper component are configured such that the additive reservoir engages the upper component with a snap fit.
8. The system of claim 1, wherein the additive reservoir is configured such that the liquid additive flows to the mixing area in response to the base liquid flowing to the spout.
9. The system of claim 1, wherein the additive reservoir includes a flexible wall.
10. The system of claim 1, wherein the additive reservoir is positioned at least partially in the inner volume when the additive reservoir is attached to the cap assembly and the cap assembly is engaged with the threaded mouth.
11. An additive delivery system for use with a portable base liquid container having a threaded mouth, an inner volume to hold a base liquid, and a cap threadedly engaged with the threaded mouth and having an opening through which the inner volume is accessible, the system comprising: an additive reservoir including at least one wall defining a space holding a liquid additive and an outlet from which the liquid additive exits the additive reservoir; a cap assembly including a spout configured for movement between open and closed positions to control flow of liquid from the spout, and an upper component to which the additive reservoir is attached and from which the additive reservoir depends, the upper component configured to threadedly engage with the cap at the opening to secure the additive reservoir to the cap and to threadedly disengage from the cap to permit removal of the upper component and the additive reservoir from the cap; and a

base liquid flow path extending through the cap assembly to the spout, the base liquid flow path fluidly coupled to receive the liquid additive from the additive reservoir at a mixing area to mix the liquid additive with the base liquid, and including a one-way valve in the base liquid flow path upstream of the mixing area that prevents backflow of the base liquid into the base liquid container, wherein the additive reservoir and the cap assembly are configured to engage with the cap such that the additive reservoir is positioned at least partially in the inner volume with the upper component threadedly engaged with the cap.

12. The system of claim 11, the cap assembly further comprising a lower component that at least partially defines an additive reservoir enclosure in which the additive reservoir is held, and a metering port configured to receive the liquid additive from the outlet of the additive reservoir and provide the liquid additive to the mixing area.

13. The system of claim 12, the lower component including an opening to permit the base liquid from the inner volume to flow into the additive reservoir enclosure.

14. The system of claim 12, wherein the lower component includes an opening to permit the base liquid from the inner volume to flow into the additive reservoir enclosure, and the base liquid that flows into the additive reservoir enclosure through the opening of the lower component does not contact the liquid additive in the additive reservoir.

15. The system of claim 11, wherein the upper component is configured to be removed from the cap while the cap is engaged with the base liquid container.

16. The system of claim 11, wherein the upper component includes a female threaded part configured to threadedly engage with a male threaded part on the cap at the opening.

17. The system of claim 11, wherein the additive reservoir and the upper component are configured such that the additive reservoir engages the upper component with a snap fit.

18. The system of claim 11, wherein the additive reservoir is configured such that the liquid additive flows to the mixing area in response to the base liquid flowing to the spout.

19. The system of claim 11, wherein the additive reservoir includes a flexible wall.
